Testing Deeper Remediation: A Within-School RCT of Personalized Adaptive Learning in India[*]

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Abstract

We report on a pre-registered within-school randomised experiment in 20 Indian government schools that tests whether deeper remediation — extending instruction up to five grade levels below a student's enrolled grade — produces larger gains in math test scores than the standard two-grade remediation depth of the Personalized Adaptive Learning (PAL) programme evaluated in our companion paper. Lab attendance and scheduled curriculum are balanced across arms; the Free Flow contrast is approximately 200 additional minutes per student over the two-year intervention of below-cap deep remediation content (grades $g-3$ through $g-5$) that the Status Quo algorithm does not render at all.

The aggregate effect on standardised math ability is small but statistically significant: $+0.053\ \sigma$ ($p = 0.016$). The implied per-minute return at this dose is approximately $0.02\ \sigma$ per hour, in the same order of magnitude as the per-hour return to total PAL exposure estimated by the IV strategy in our companion paper. The pooled effect is broadly distributed across enrolled grades — positive at Grade 6 ($+0.11\ \sigma$), Grade 7 ($+0.06\ \sigma$), and Grade 9 ($+0.03\ \sigma$); near zero at Grade 8 — with no detectable heterogeneity by pre-PAL math ability. Decomposing the endline by content grade of the test item shows the strongest gain on shallow-remediation items ($+0.072\ \sigma$, $p = 0.001$ pooled; $+0.164\ \sigma$, $p = 0.006$ in Grade 6), with the pooled deep-remediation effect intermediate ($+0.051\ \sigma$, $p = 0.018$) and no detectable effect on at-grade items — consistent with deep-content exposure producing transferable prerequisite learning at content depths one to two grades above where the distinctive dose lands.

JEL Codes: I21, I25, O15, C93

Keywords: Personalized Adaptive Learning, Remediation, EdTech, India

[*]Companion paper to Cupito et al. (2026), "Can the Public Sector Deliver Adaptive Learning Technology at Scale? Experimental Evidence from India." All affiliations as in the companion paper.

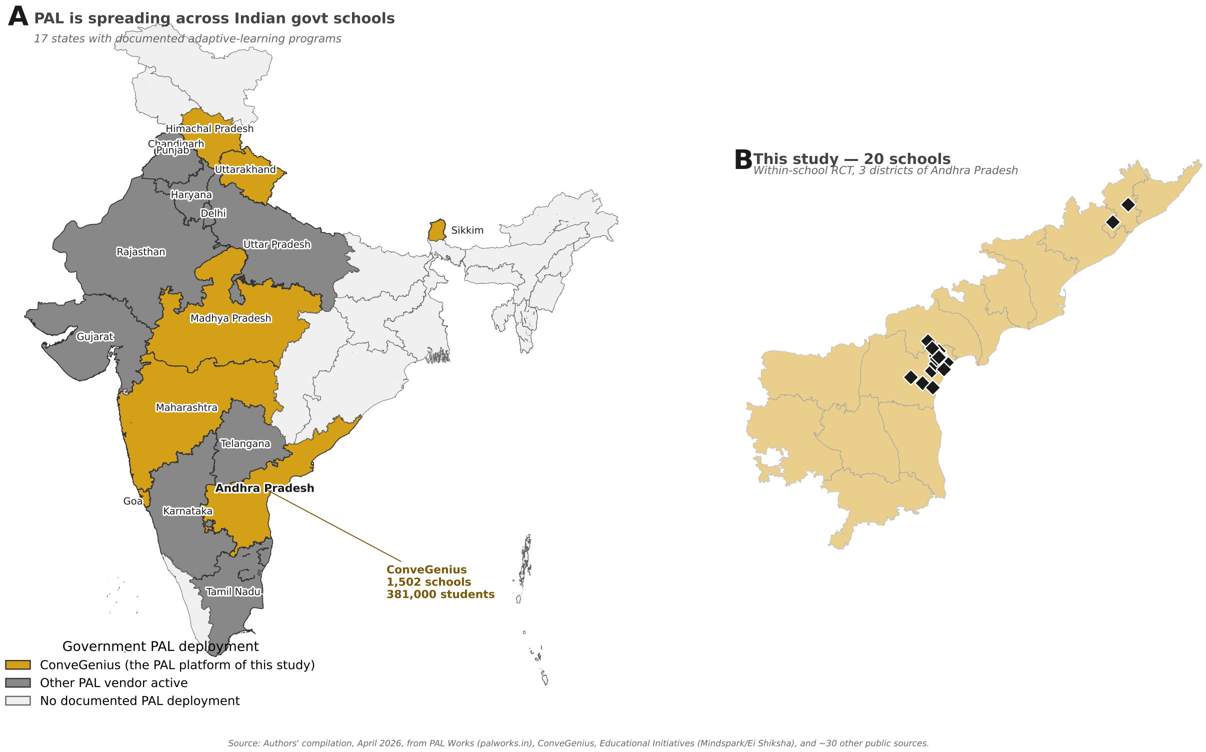

FIGURE 1. PAL ADOPTION ACROSS INDIAN GOVERNMENT SCHOOLS, WITH THE 20 REMEDIATION-STUDY SCHOOLS

Notes: **Panel A:** Indian states colour-coded by PAL deployment status. Gold = states where ConveGenius (the platform behind this study) operates: Andhra Pradesh, Himachal Pradesh, Madhya Pradesh, Maharashtra, Goa, Uttarakhand, Sikkim. Dark grey = states where another adaptive-learning vendor is documented in government schools (Mindspark/Educational Initiatives, Khan Academy India, iDream Education, Schoolnet/Geneo, Chimple, Eduten). Light grey = no documented government PAL deployment. The annotation for Andhra Pradesh shows the scale of the underlying programme (1,502 schools, 381,000 students; *The Week*, January 2026). **Panel B:** The 20 schools of this within-school remediation RCT, distributed across 3 districts of Andhra Pradesh (Bapatla, Prakasam, Vizianagaram); randomisation to Free Flow / Status Quo arms is at the student level within each school. State-level PAL coverage data compiled April 2026 from public sources; full data and source URLs in `code_claude/00_project_knowledge/pal_india_state_coverage.csv`. The Avanti Fellows / Pragati / Gurukul programme is excluded as a JEE/NEET test-prep platform with light personalisation, not adaptive learning in the strict PAL sense.

I. Summary

The Government of Andhra Pradesh's Personalized Adaptive Learning (PAL) programme, evaluated in our companion paper, raises endline math ability by $+0.39\ \sigma$ across the 60 treatment schools of a 120-school randomised trial. PAL operates a curriculum-aligned tablet platform that diagnoses each student's current level and renders content appropriate to that level, with remediation extending up to two grade levels below the student's enrolled grade. This two-grade cap is partly a pedagogical choice and partly an accommodation to a political-economy constraint: serving multi-grade-below content to enrolled-grade students makes visible to teachers, parents, and administrators that a substantial share of students are operating several grades below curricular expectation. The cap leaves open the question this paper addresses: would deeper remediation produce additional learning gains?

We test this through a pre-registered within-school randomised experiment in 20 schools drawn from the high-usage tail of the AP programme, distributed across three districts (Bapatla, Prakasam, Vizianagaram). Within each school, students in grades 6 through 9 are individually assigned to one of two PAL algorithm configurations: Status Quo, which caps remediation at two grades below enrolled grade, or Free Flow, which extends remediation up to five grades below. Lab sessions are scheduled by the classroom teacher at the school-grade level and teachers are not informed of arm assignment, so scheduled exposure to PAL is identical across arms by design. The experiment isolates the marginal learning return to access to deeper content, holding scheduled exposure and curriculum delivery fixed.

Section II documents that the experimental contrast is in fact the content-depth contrast the design intended. Lab attendance is full and equal: median session start and end times in (school $\times$ grade $\times$ day) cells align within minutes between arms. The PAL algorithm covers the same chapter set in both arms at at-grade content (Jaccard 0.99) and at shallow remediation (Jaccard 0.85, with Status Quo's set 95% contained within Free Flow's). The behavioural contrast is concentrated at content depths the Status Quo algorithm does not render: Free Flow students log $+197$ minutes per student of below-cap remediation material ($g-3$ through $g-5$) over the two-year intervention, accounting for roughly 90% of the total Free Flow$-$Status Quo engagement gap of $+218$ minutes. The intent-to-treat contrast is therefore essentially equivalent to the treatment-on-treated effect of access to approximately 200 additional minutes of deep-remediation content.

The aggregate learning effect on standardised math ability is small but statistically significant: $+0.053\ \sigma$ ($p = 0.016$, Section IV). The implied per-minute return at this dose is approximately $0.02\ \sigma$ per hour, in the same order of magnitude as the per-hour return to total PAL exposure estimated by the instrumental variables strategy in our companion paper. The aggregate is broadly distributed across enrolled grades — positive at Grade 6 ($+0.11\ \sigma$), Grade 7 ($+0.06\ \sigma$), and Grade 9 ($+0.03\ \sigma$), and near zero at Grade 8 — and is not detectably heterogeneous by pre-PAL math ability: the joint F-test of equal Free Flow effects across within-school-by-grade ability terciles is far from rejection ($F = 0.12$, $p = 0.88$, Section V).

Decomposing the endline by content grade of the test item locates where the gain materialises (Section VI). Pooling across enrolled grades, the Free Flow effect is largest on shallow-remediation items ($+0.072\ \sigma$, $p = 0.001$ on items at content grades $g-1$ and $g-2$), intermediate on deep-remediation items ($+0.051\ \sigma$, $p = 0.018$ on items at $g-3$ and below), and undetectable on at-grade items ($+0.016\ \sigma$, $p = 0.56$). The pattern is sharpest at Grade 6, where Free Flow students gain $+0.164\ \sigma$ ($p = 0.006$) on shallow-remediation items — improvement one to two content-grades *above* where the distinctive deep-content dose was rendered. The spill-up pattern is consistent with deep content building prerequisite skills — basic arithmetic, place value, whole-number operations — that transfer to Grade-4 and Grade-5 material on which students sit closer to a competency threshold.

The aggregate effect is small. Whether this reflects a low marginal return to deeper remediation or a treatment dose too modest to produce a larger detectable effect cannot be distinguished from a single experiment at this scale. Two features of the design bound interpretation. Statistical power is concentrated in pooled tests; tercile- and grade-level effects are estimated from $\sim$ 1,150 and $\sim$ 830–1,050 students respectively. And the twenty-school sample is drawn from the high-usage tail of the AP programme; in schools with weaker implementation, both the responding subset and the dose of deep-remediation engagement Free Flow induces may differ.

The remainder of the paper is organised as follows. Section II characterises the experimental contrast — attendance, curriculum, and content-depth dose. Section III describes the analysis sample, the IRT-scored math endline, the content-based treatment definition, and the estimating equation. Section IV reports the intent-to-treat effect on standardised math ability. Section V tests whether the effect varies with pre-PAL baseline ability. Section VI decomposes the effect by content grade of the test item. Section VII discusses.

II. Intervention

Randomisation is at the student-within-school level: within each of the 20 study schools, students in grades 6–9 are individually randomised to Free Flow (`GOAPFF_1023`, remediation depth unlimited) or Status Quo (`GOAPM_0823`, remediation capped at 2 grade levels below enrolled grade). PAL lab sessions are scheduled by classroom teachers at the school-grade level; teachers are not informed of arm assignment. Baseline scheduled exposure to PAL — lab days, lab start times, lab lengths — should therefore be identical across arms. The remainder of this section documents that scheduled exposure is indeed equal, that the PAL algorithm exposes the same curriculum to both arms, and that the treatment contrast manifests entirely as a difference in *content depth*: access to below-cap remediation material that the Status Quo algorithm does not render.

Physical lab attendance is identical across arms. Figure 2 shows the weekly session pattern across all 20 schools: at every school $\times$ day-of-week cell, the Free Flow (navy) and Status Quo (grey) session bars lie on top of one another. Across 4,666 (school $\times$ grade $\times$ day) cells in which at least three students from each arm were active over the two-year intervention (AY 2023–24 and 2024–25 pooled, with December 2023 and January 2024 dropped from both arms because Free Flow data for those months is incomplete; see footnote 1), the median difference between arms in session start and end time is essentially zero, and the cells align within ± 10 min on both edges in roughly four out of five cases (Appendix Figure 6). Physical compliance is full and equal.[1]

Conditional on identical attendance, we examine whether the PAL algorithm renders the same instructional material to each arm. For each (school $\times$ grade) class, we compute the set of distinct content-chapters each arm's students touched over the two-year intervention (excluding the December 2023 and January 2024 gap window), keyed by (content-grade, chapter number) so that Telugu-medium and English-medium renderings of the same chapter resolve to a single identifier. Table 1 reports the arm-level overlap by content-grade. At at-grade content, the Free Flow and Status Quo chapter sets overlap at Jaccard 0.99 — the two arms cover essentially identical curricula. Overlap remains high at shallow remediation (Jaccard 0.85), where Status Quo's chapter set is 95% contained within Free Flow's; Free Flow students touch a small handful of additional chapters per class but never skip content Status Quo visits. At deep remediation, virtually 100% of Status Quo's (small) chapter footprint is a subset of Free Flow's, and Free Flow students access on average 3.3 additional chapters per class that the Status Quo algorithm does not render. The PAL algorithm is content-conservative: the material exposed to Status Quo is always a subset of what Free Flow students see.

Conditional on identical attendance and overlapping curricula, Free Flow students log modestly more total engagement than Status Quo students. Across the bilateral two-year intervention window, Free Flow students average 20.2 hours of wall-clock PAL engagement per student versus 17.9 for Status Quo (a $+2.3$-hour Free Flow advantage); the in-app `total_time_spent_new` timer, which accumulates pause and background time within each open module and therefore exceeds wall-clock by a small multiplicative factor, registers a $+3.7$-hour Free Flow advantage (21.2 versus 17.6 hours, a $+21\%$ gap). The in-app gap decomposes multiplicatively into three factors of similar magnitude, shown in Figure 3: Free Flow students are active on 9% more days per student (28.2 versus 25.9), complete 3% more modules per active day (25.7 versus 24.9), and each of their modules has a 9% longer in-app timer (1.85 versus 1.70 min). Compounded, $1.09 \times 1.03 \times 1.09 \approx 1.22$,

[1] The CG re-export delivered on 2026-04-27 cleaned four GOAPM (Status Quo) files but did not include re-exports of the GOAPFF (Free Flow) data for December 2023 (entirely missing) or January 2024 (partial, $\approx$25% of expected rows). Including those months in their current state would bias the FF-vs-SQ comparison against Free Flow. We therefore drop December 2023 and January 2024 from both arms throughout the intervention-section analyses, retaining bilateral parity.

these factors reproduce the observed 21% in-app gap.

The engagement gap also distributes unevenly across CG module types. Figure 4 reports per-student minutes by module type. The bulk of the Free Flow excess lies in Video modules (+238 minutes per student over the two-year intervention, a 42% increment) and in Remedial Practice modules (+22 minutes, 40%). Formative assessments — the second-largest module-type bucket overall — are essentially balanced across arms (+12 minutes on a 271-minute base, 4%). The other four module types (Chapter Practice, Baseline, Diagnostic, Endline) contribute negligibly to the gap. The algorithmic FF–SQ contrast therefore operates at the module-type dimension as well as the content-grade dimension we turn to next.

What separates the arms substantively, however, is where that engagement lands in content space. Figure 5 allocates each arm's logged PAL minutes across content-grade relative to enrolled grade. At $g-3$ through $g-5$ — the Status Quo cap frontier and below — Free Flow students register an additional +197 minutes per student over the two-year intervention that Status Quo students cannot access. This below-cap differential accounts for $\approx 90\%$ of the total Free Flow–Status Quo engagement gap of +218 minutes; the remaining 10% appears as a small Free Flow excess at at-grade content (+22 minutes), with shallow remediation ($g-1$ and $g-2$) essentially balanced (-5 at $g-2$, +6 at $g-1$). The Free Flow configuration's distinctive treatment is therefore additional below-cap remediation content that the Status Quo algorithm does not render at all. The below-grade dose differential is positive and significant at every grade: Grade 6 +3.3 hours, Grade 7 +3.1 hours, Grade 8 +3.2 hours, Grade 9 +3.6 hours over the bilateral two-year intervention window (all $p < 0.001$). Appendix Figure 7 reports the full grade-wise breakdown.

Taken together, lab attendance is full and equal; the PAL algorithm exposes the same curriculum, in the same order, to both arms; and the behavioural Free Flow excess of ≈ 218 minutes per student over the two-year intervention is concentrated at content depths the Status Quo algorithm does not render. We therefore treat the intent-to-treat contrast as essentially equivalent to the treatment-on-treated: the causal margin we estimate in the sections that follow is the math-learning effect of access to deeper remediation content, delivered at approximately 200 additional minutes (3.3 hours) of below-cap engagement per student across the two-year intervention.

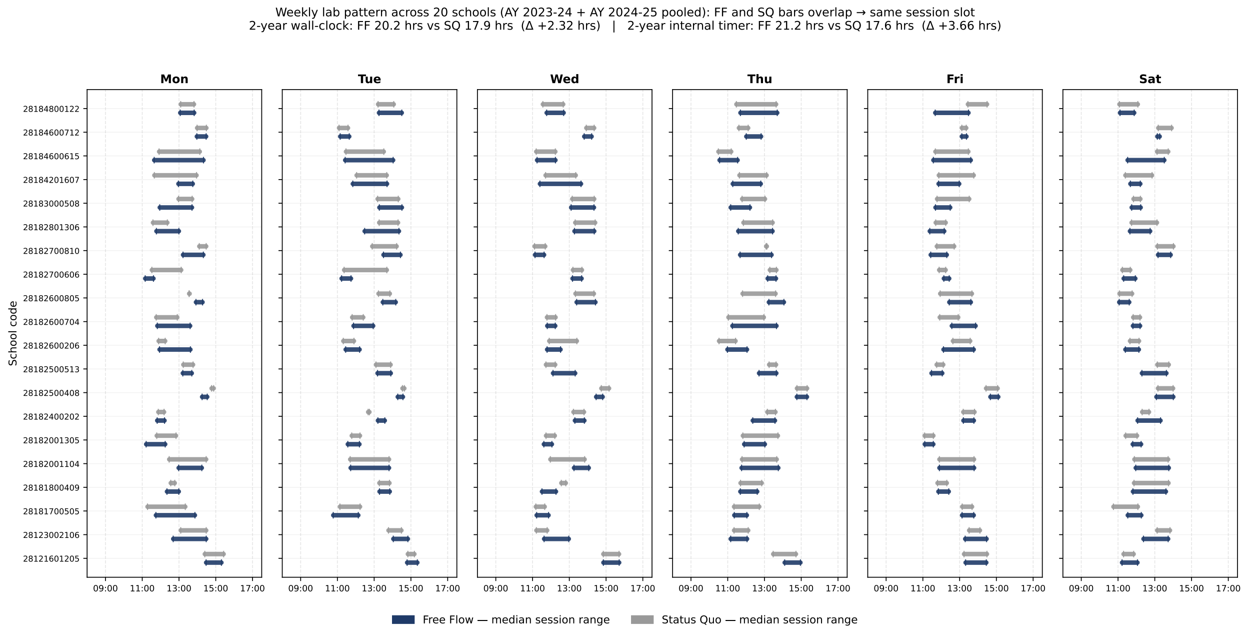

FIGURE 2. FREE FLOW AND STATUS QUO STUDENTS ATTEND THE SAME LAB SESSIONS

Notes: Weekly PAL lab-session pattern across all 20 remediation-study schools, Monday through Saturday, AY 2023–24 + AY 2024–25 pooled (December 2023 and January 2024 dropped from both arms; see footnote 1). For each (school × day-of-week) cell with at least three qualifying session-days across the two years, horizontal bars show the median session start-to-end time separately for Free Flow (navy) and Status Quo (grey) students. Bars lie on top of one another at every cell: arms share the same teacher-scheduled lab slots. Two-year engagement comparison (top of figure): wall-clock PAL engagement is Free Flow 20.2 hours versus Status Quo 17.9 hours ($\Delta = +2.3$ hours); in-app timer minutes register 21.2 versus 17.6 ($\Delta = +3.7$ hours), with the wedge between wall-clock and in-app timer reflecting module-timer inflation over wall-clock. Lab-session ranges computed per-student from first to last module submission on a given day; cell medians taken across students within each arm. See Appendix Figure 6 for the signed start/end delta across the 4,666 (school × grade × day) cells that qualify.

TABLE 1—FREE FLOW AND STATUS QUO STUDENTS COVER THE SAME CHAPTERS

Content level	**Jaccard** $\lvert FF \cap SQ \rvert / \lvert FF \cup SQ \rvert$	**FF ⊆ SQ** % of FF ch.	**SQ ⊆ FF** % of SQ ch.	**FF / SQ / shared** mean # chapters
At-grade (g)	0.992	99.6%	99.6%	14.3 / 14.3 / 14.3
Shallow remediation ($g-1$, $g-2$)	0.848	88.4%	95.3%	17.6 / 16.2 / 15.4
Deep remediation ($g-3$, $g-4$, $g-5$)	0.614	61.5%	99.7%	8.0 / 4.7 / 4.7

Notes: Per (school × enrolled-grade) class ($N = 80$ classes with ≥ 5 common chapters in the at-grade and shallow-remediation rows; $N = 44$ in the deep-remediation row), the set of distinct content-chapters touched by Free Flow students and by Status Quo students over the two-year intervention (AY 2023–24 + AY 2024–25), by content-grade relative to enrolled grade. "FF ⊆ SQ" reports the share of Free Flow's chapter set also covered by Status Quo; "SQ ⊆ FF" reports the reverse. Chapters are keyed by (content-grade, chapter-number) so that Telugu-medium and English-medium renderings of the same chapter resolve to a single identifier. Shallow-remediation: Status Quo's set is 95% contained in Free Flow's; deep-remediation: Status Quo's set is 99.7% contained in Free Flow's but Free Flow additionally covers 3.3 chapters per class that the Status Quo algorithm does not render.

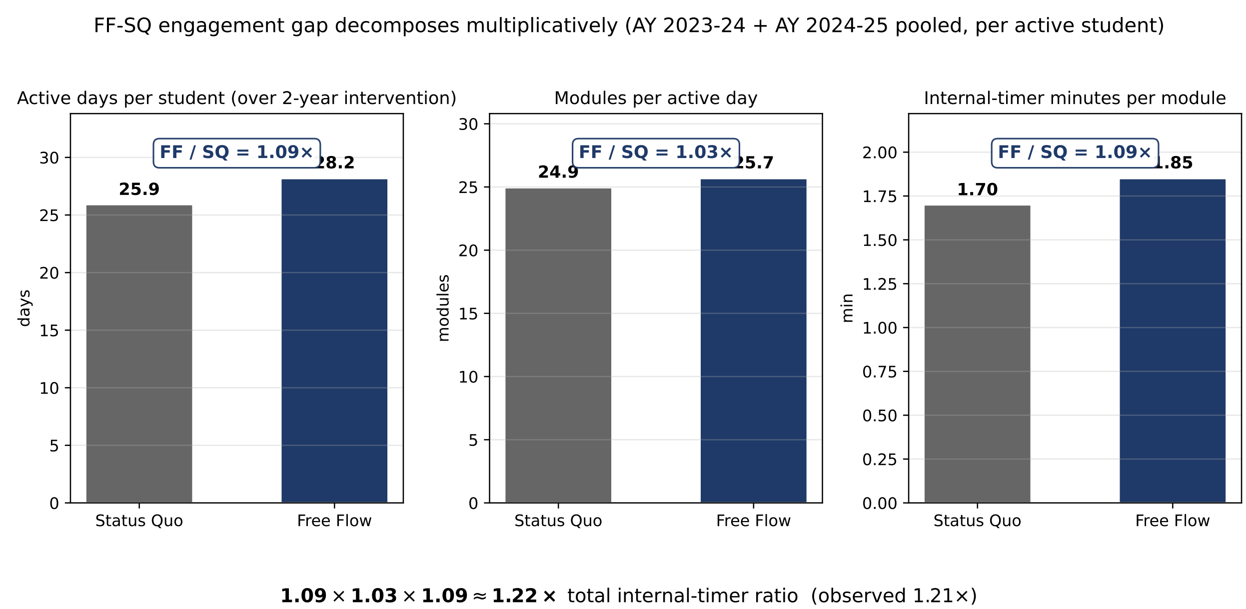

FIGURE 3. THE FREE FLOW–STATUS QUO ENGAGEMENT GAP DECOMPOSES MULTIPLICATIVELY

Notes: Per active student, AY 2023–24 + AY 2024–25 pooled (December 2023 and January 2024 dropped from both arms; see footnote 1). The Free Flow–Status Quo in-app timer gap of +21% decomposes into three multiplicative factors: Free Flow students are active on 9% more days per student (28.2 versus 25.9), complete 3% more modules per active day (25.7 versus 24.9), and each of their modules has a 9% longer in-app timer (1.85 versus 1.70 min). Compounded, $1.09 \times 1.03 \times 1.09 \approx 1.22$. Active days and modules per day are measured from timestamped CG module submissions; in-app timer minutes are the CG platform's internal per-module timer (`total_time_spent_new`), which accumulates pause and background time within a module.

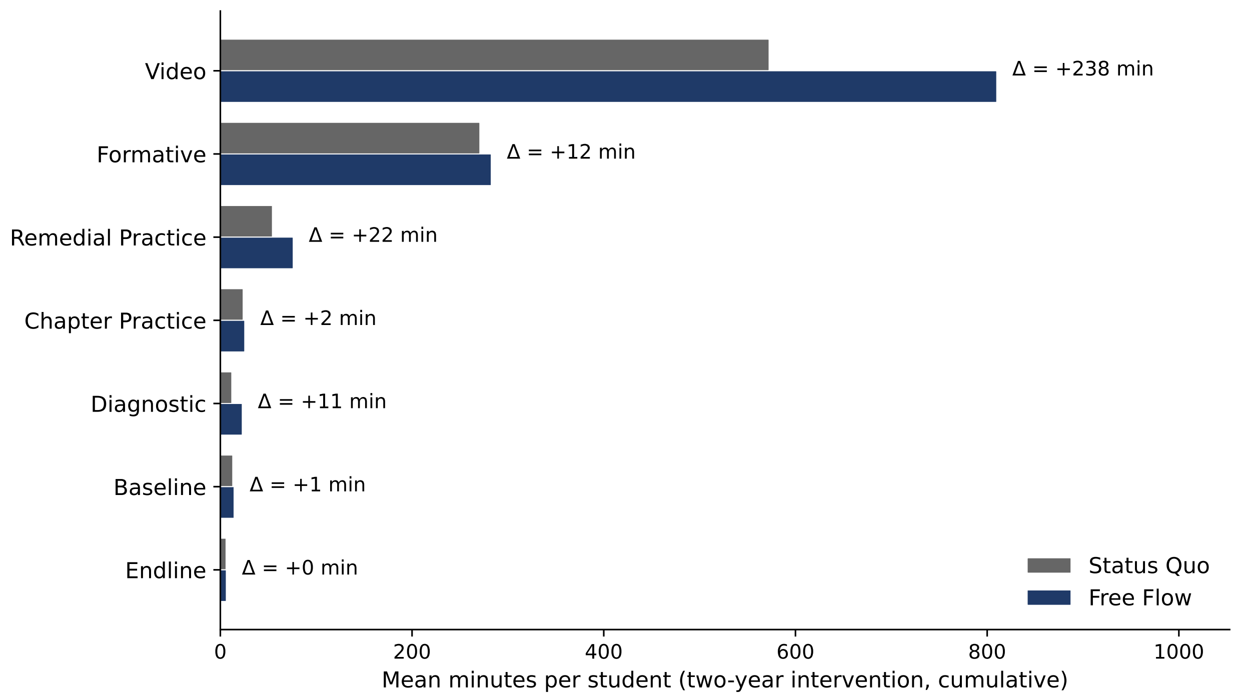

FIGURE 4. THE FREE FLOW–STATUS QUO ENGAGEMENT GAP CONCENTRATES IN VIDEO AND REMEDIAL PRACTICE MODULES

Notes: Mean minutes per student of PAL engagement, two-year cumulative (AY 2023–24 + AY 2024–25), broken down by CG `module_type`. Module types are sorted by total volume across arms. The Free Flow excess of $+218$ minutes per student concentrates in Video ($\Delta = +238$ min, $+42\%$ over Status Quo) and in Remedial Practice ($\Delta = +22$ min, $+40\%$); Formative assessments — the second-largest module-type bucket overall — are essentially balanced across arms ($\Delta = +12$ min on a 271-min base, $+4\%$). Diagnostic, Chapter Practice, Baseline, and Endline modules account for negligibly small per-student volumes and contribute little to the FF–SQ gap.

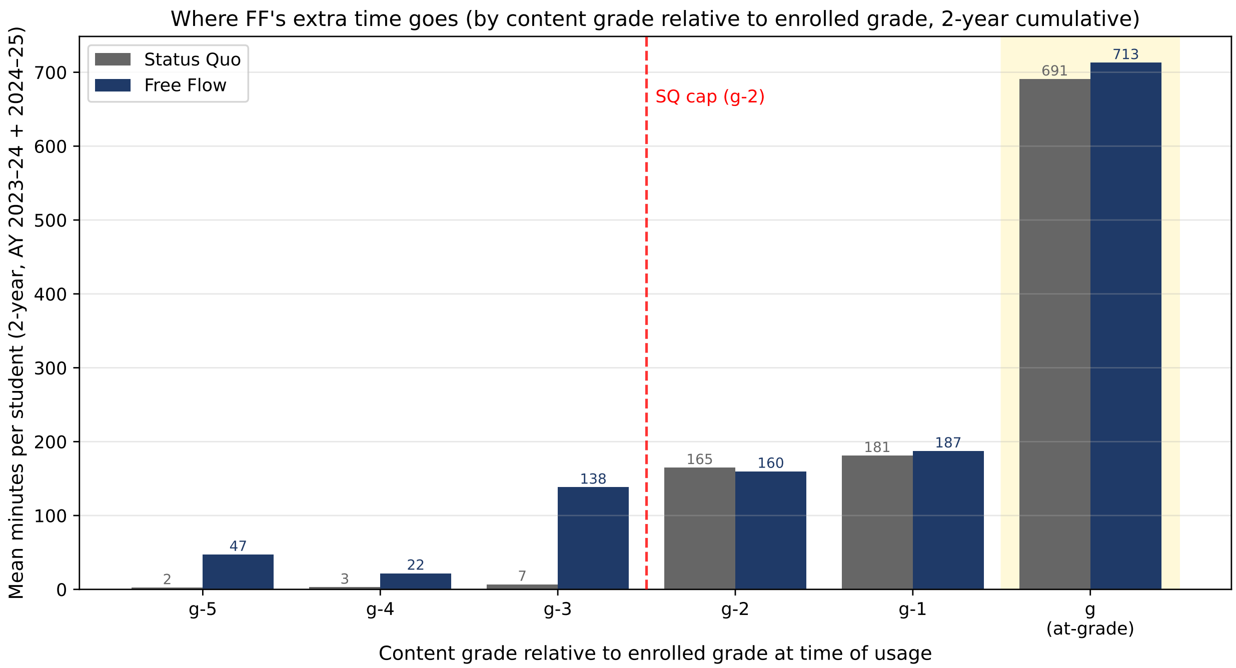

FIGURE 5. THE FREE FLOW–STATUS QUO ENGAGEMENT GAP LIVES ENTIRELY BELOW THE STATUS QUO CAP

Notes: Mean minutes per student of PAL engagement, two-year cumulative (AY 2023–24 + AY 2024–25, December 2023 and January 2024 dropped from both arms; see footnote 1), by content-grade relative to enrolled grade. At $g-3$ through $g-5$ — below the 2-grade Status Quo cap — Free Flow students register an additional $+197$ minutes per student that Status Quo students cannot access. At at-grade (g), Free Flow has a small advantage ($+22$ minutes); at shallow remediation ($g-1$ and $g-2$) the arms are essentially balanced ($+6$ at $g-1$, -5 at $g-2$). The below-cap differential accounts for $\approx 90\%$ of the total Free Flow–Status Quo engagement gap of $+218$ minutes. The figure characterises the intensive margin of the treatment: at how deep a remediation level the algorithm exposes each arm's students to material.

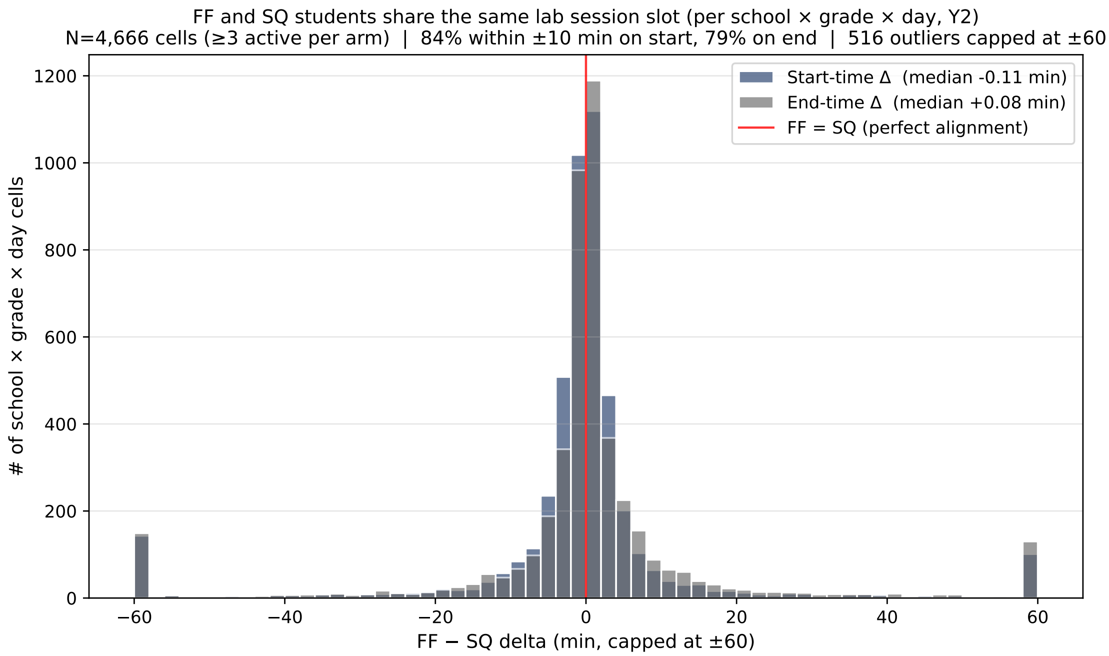

FIGURE 6. FF AND SQ SESSION SLOTS ALIGN WITHIN MINUTES AT THE SCHOOL-GRADE-DAY LEVEL

Notes: For each (school × grade × day) cell with at least three active students in each arm (4,666 cells across the two-year intervention), we compute the median first-submission time (navy) and median last-submission time (grey) separately per arm, and take the difference FF − SQ. Both signed-delta distributions are tightly concentrated at zero; the median delta is essentially zero on both start and end times. Roughly four out of five cells align within ±10 min on both start and end. Edge bars at ±60 min represent the ∼5–6% of cells with misaligned sessions (distinct lab periods scheduled on the same day) — these cells are retained in the session-slot alignment claim because the median delta, rather than the mean, is reported.

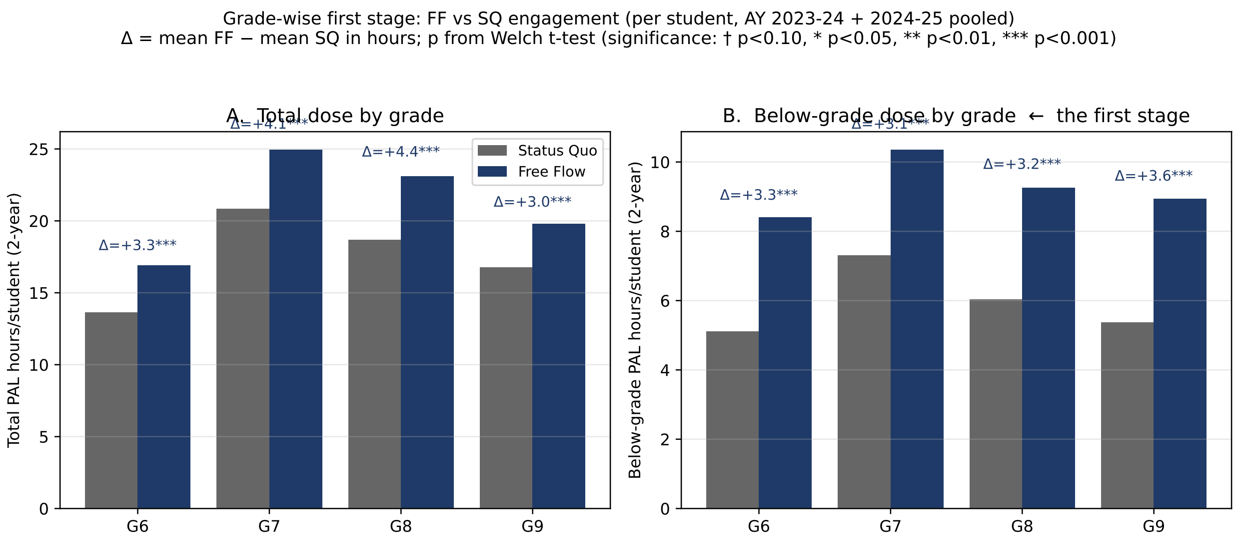

FIGURE 7. THE FREE FLOW FIRST STAGE BY ENROLLED GRADE

Notes: Mean PAL hours per student over the two-year intervention (AY 2023–24 + AY 2024–25, December 2023 and January 2024 dropped from both arms; see footnote 1), by enrolled grade × arm. Panel A: total dose. Panel B: below-grade dose — the cap differential. The Free Flow algorithm exposes students to remediation up to five grade levels below enrolled grade; the Status Quo algorithm caps at two. Δ = mean FF − mean SQ in hours; significance from two-sided Welch t-test († $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). All four grade contrasts are positive and significant on both total and below-grade dose: total dose Free Flow − Status Quo is +3.3, +4.1, +4.5, and +3.0 hours at Grades 6 through 9 respectively; below-grade dose differential is +3.3, +3.1, +3.2, and +3.6 hours. The below-grade dose differential is roughly the same size as the total dose differential at each grade, indicating that Free Flow's extra time is concentrated at content depths the Status Quo algorithm cannot render.

III. Data and Empirical Design

Our analysis sample consists of the 3,755 students who completed the endline assessment in February–March 2025 in the 20 remediation-study schools and for whom treatment exposure can be determined from ConveGenius (CG) module-level records. Only one of the 3,756 endline respondents in the master roster is unclassifiable: a student with neither a backend roster record (in either DIL or CG) nor any PAL usage at all. The 3,755 enter every regression.

We define the treatment indicator FF_i from *content actually rendered* by the algorithm to each student over the two-year intervention, rather than from the backend randomisation assignment:

$$\mathrm{FF}_i \;=\; 1 \;\text{ iff }\; \text{student } i \text{ logged any module at content-grade} \;\leq\; \text{enrolled-grade} - 3,$$

i.e., received any *deep-remediation* content (three or more grade levels below enrolled grade) at any point in the intervention. The Status Quo PAL algorithm strictly caps at two grade levels below enrolled grade; in our 2.5 million CG module-level records, no module rendered by the Status Quo algorithm has content-grade below $g-2$. Accordingly, any positive deep-remediation exposure proves that the student was served by the Free Flow algorithm during at least some module sessions, while zero deep-remediation exposure indicates the student was served exclusively by the Status Quo algorithm. The treatment indicator inherits the cleanness of this algorithmic dichotomy and is invariant to whatever administrative relabelling occurred between backend assignment and module-level delivery.

This definition departs from a strict intent-to-treat specification based on the original DIL randomisation. We adopt it because backend randomisation and CG-delivered arm disagree for 14% of endline respondents (448 of 3,209 students with both records). Investigation of these mismatches reveals that they reflect within-experiment program switches: for 99% of mismatch students with both years of CG records (307 of 310), CG's algorithm placed them in one arm in 2023–24 and the other arm in 2024–25, so neither year's modal program nor the backend record fully captures their actual exposure. For such cross-over students, the depth-of-content signal is the cleanest single summary of treatment receipt. As a side benefit, the content-based definition recovers 364 endline respondents who would be dropped under strict DIL ITT due to missing or invalid backend records but who do appear in CG module-level data with a determinable arm.

Table 2 reports grade-by-arm sample sizes together with demographic descriptives. The sample is 47%

female, averages 12.4 years of age, and is drawn from AP's reserved-category population: 35% are Scheduled Tribe, 16% Scheduled Caste, and 48% Backward Class (the bulk of the rest). Father's education averages 2.2 and mother's 2.0 on a 0–5 scale, with the modal family at Grade 8 schooling. The grade composition (Grade 6 to 9) is slightly top-heavy (28% in Grade 9, 22% in Grade 6); we report all analyses both pooled and by-grade.

The primary outcome is math ability measured by a 72-item Math Master Challenge (MMC) administered at school in February–March 2025. The MMC is the same tablet-delivered instrument used in our companion paper's main-impact evaluation; each student receives a grade-adaptive subset of items spanning their enrolled grade and lower grades across the five PAL topics (algebra, geometry, statistics, number systems, measurement). We fit a two-parameter logistic IRT model jointly on the union of the 3,756-student remediation endline sample and the 14,215-student governance RCT endline sample, $\approx$ 17,971 students in total, recovering joint item parameters (a,b). Each student's ability estimate θ is then computed using these joint item parameters, so the latent ability scale is common across the remediation and governance papers. We standardise θ to the Status Quo within-school control group's mean and standard deviation and denote the standardised outcome θ^{std}. Reported coefficients remain in "σ above the SQ-control mean" units, and the underlying θ is comparable in scale to the companion paper's θ.

Our pre-specified covariates are: age, female, four caste dummies, an SES index constructed as the first principal component of nine household assets, and father's and mother's education each on a 0–5 scale. We report pre-randomisation balance on these covariates and on a cohort-specific pre-PAL baseline math score (Grade 6 endline cohort: AY 2023–24 SA1, taken in their Grade 5 year before PAL; Grades 7–9 endline cohorts: AY 2022–23 SA1, their Grade 5–7 year, also pre-PAL), but do not include the baseline math score as a regression covariate in the main specifications — coverage is 93%, and the contemporaneous AY 2023–24 SA1 used in earlier drafts was administered in mid-2024, after the Grade 7–9 endline cohorts had already begun PAL in their G6/G7/G8 year, so it carries part of the Year-1 PAL treatment effect. Missing covariate values are imputed with the sample mean (continuous variables) or the omitted category (dummy variables) and accompanied by a missing-indicator dummy, preserving the full classified sample of 3,755 students.

The estimating equation is

$$\theta_{isg}^{\text{std}} = \alpha_{sg} + \beta\,\text{FF}_{isg} + \mathbf{X}'_{isg}\gamma + \varepsilon_{isg},$$

where i indexes students, s schools, and g grades; FF is the content-based treatment indicator defined above; $\mathbf{X}_{isg}$ is the covariate vector above; and α_{sg} is a school-by-grade-by-section fixed effect, absorbing the original DIL randomisation stratum (school $\times$ grade $\times$ section). The section variable used for randomisation in 2023 is not preserved in the DIL backend output; we recover it from the AY 2023–24 government enrollment roster (the input file Sharad Hotha's randomisation script operated on) and, for the Grade 6 cohort that joined in AY 2024–25, from the AY 2024–25 government enrollment file. Together these recover section for 98.4% of randomised students; the residual is folded into a within-school $\times$ grade "unknown" cell. Standard errors are clustered at the school level ($S = 20$).

Table 3 reports baseline balance under this content-based classification. Caste shares, age, parental education, and the SES index are tightly balanced across arms, but Free Flow students are slightly more likely to be female (+6 percentage points, $p = 0.067$), and the joint F-test on the nine covariates is marginally significant ($F = 2.63$, $p = 0.040$). Pre-PAL math, constructed cohort-specifically (Grade 6 endline $\rightarrow$ AY 2023–24 SA1, taken in their Grade 5 year before PAL; Grades 7–9 endline $\rightarrow$ AY 2022–23 SA1, their Grade 5–7 year, also pre-PAL), is essentially balanced (-0.28 points, $p = 0.61$). The remaining mild covariate-vector imbalance is the price of the as-rendered treatment definition: classification depends on actual algorithm-engagement, which correlates with student characteristics. Because pre-PAL baseline math is balanced and the only available pre-PAL test score covers only 93% of the sample, we do not include baseline math as a regression covariate; including it as a control, particularly the contemporaneous AY 2023–24 SA1 used in earlier drafts, would absorb part of the Year-1 PAL effect rather than pre-randomisation heterogeneity for the Grade 7–9 cohorts (whose AY 2023–24 SA1 was administered after they had begun PAL).

TABLE 2—SAMPLE CHARACTERISTICS BY GRADE AND ARM

	N		Female		Age (yrs)		SES index		Baseline math	
	SQ	FF	SQ	FF	SQ	FF	SQ	FF	SQ	FF
Grade 6	399	434	0.489	0.527	10.76	10.73	0.14	0.02	32.0	32.1
Grade 7	423	460	0.430	0.490	11.92	11.81	-0.05	-0.02	32.2	33.2
Grade 8	448	536	0.431	0.456	12.82	12.83	-0.01	-0.03	28.9	26.6
Grade 9	469	587	0.408	0.498	13.88	13.92	0.01	-0.03	36.7	35.0
All grades	1,739	2,017	0.438	0.491	12.42	12.46	0.02	-0.02	32.5	31.7

Notes: Means of student characteristics for the 3,755-student analysis sample, separately by enrolled grade (rows) and arm (columns). Arm assignment is the content-based treatment indicator defined in Section III: a student is classified Free Flow if and only if they received any deep-remediation content ($\leq g-3$) from the algorithm. N reports student count; female, age, and SES index are drawn from the demographic survey, and baseline math is the cohort-specific pre-PAL government Summative Assessment 1 (Grade 6 endline cohort: AY 2023–24; Grades 7–9 endline cohorts: AY 2022–23). Pre-PAL baseline math is missing for approximately 7% of the sample.

TABLE 3—COVARIATE BALANCE BETWEEN FREE FLOW AND STATUS QUO ARMS

	Status Quo (1)	Free Flow (2)	Difference (3)	(SE) (4)	p-value (5)	(
Female	0.438	0.491	0.057*	(0.031)	0.067	3,7
Age (years)	12.42	12.46	-0.017	(0.035)	0.624	3,6
Scheduled tribe	0.365	0.342	0.007	(0.015)	0.650	3,6
Scheduled caste	0.156	0.168	0.010	(0.011)	0.345	3,6
Backward class–A	0.048	0.039	-0.010	(0.007)	0.157	3,6
Backward class–B	0.432	0.451	-0.007	(0.012)	0.545	3,6
Father's education (0–5)	2.21	2.20	-0.047	(0.063)	0.456	3,4
Mother's education (0–5)	1.95	1.98	0.027	(0.067)	0.688	3,3
SES index (PCA)	0.02	-0.02	-0.026	(0.030)	0.394	3,7
Joint F-statistic			2.63		$[p = 0.040]$	3,3
Pre-PAL baseline math (cohort-specific govt SA1)	32.50	31.72	-0.282	(0.548)	0.606	3,4

Notes: Means of each covariate by arm (columns 1–2) and the estimated Free Flow − Status Quo difference (column 3) from OLS of the covariate on Free Flow plus school-by-grade-by-section fixed effects. Standard errors clustered at the school level ($S = 20$). Caste dummies follow the Andhra Pradesh reservation categories; Backward Class B is the omitted reference in the joint F-test to avoid collinearity with the other three caste indicators. Pre-PAL baseline math is constructed cohort-specifically: AY 2023–24 SA1 for the Grade 6 endline cohort (their Grade 5 year, before PAL); AY 2022–23 SA1 for the Grade 7–9 endline cohorts (their Grade 5–7 year, also pre-PAL). Coverage is 93% of the analysis sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IV. Main Results

Table 4 reports estimates of the Free Flow arm's effect on standardised math ability (θ^{std}) from the specification of Equation 1, pooled across grades and separately for each enrolled grade.

The pooled Free Flow effect is small but statistically significant: $+0.053\ \sigma$ ($p = 0.016$). The magnitude is small against the backdrop of the intervention contrast documented in Section II: Free Flow students

TABLE 4—FREE FLOW EFFECT ON STANDARDISED MATH ABILITY (ITT)

	All grades	Grade 6	Grade 7	Grade 8	Grade 9
	(1)	(2)	(3)	(4)	(5)
Free Flow	+0.053**	+0.113*	+0.059	+0.016	+0.031
	(0.022)	(0.063)	(0.057)	(0.050)	(0.066)
Status Quo mean	+0.000	-0.310	+0.040	+0.049	+0.181
Observations	3,755	832	883	984	1,056
R^2	0.231	0.249	0.263	0.227	0.172

Notes: Intent-to-treat estimates of Free Flow on standardised math ability (θ^{std}), pooled across grades (column 1) and separately by enrolled grade (columns 2–5). Specification: OLS of θ^{std} on a Free Flow indicator, a vector of pre-specified covariates, and school-by-grade-by-section fixed effects; standard errors clustered at the school level ($S = 20$). Covariates: age, female, father's and mother's education (dummies for each level), SES index, with missing-indicator imputation. θ^{std} is standardised to the Status Quo control group's mean and standard deviation within the remediation sample; the "Status Quo mean" row reports that group's mean within each column's sample and therefore deviates from zero in the by-grade columns. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

receive an additional $\approx$ 200 minutes ($\approx$ 3.3 hours) of below-cap remediation material across grades $g-3$ to $g-5$ over the two-year intervention, a content-depth margin that the Status Quo algorithm does not render at all. Exposing students to that deeper remediation content produces a detectable but modest additional learning gain above the Status Quo two-grade cap.

The pooled estimate is broadly consistent across enrolled grades. Point estimates are positive at three of four grades — Grade 6 (+0.113 σ, $p = 0.075$), Grade 7 (+0.059 σ), and Grade 9 (+0.031 σ) — and essentially zero at Grade 8 (+0.016 σ). The Grade 6 estimate is the largest in absolute value and the only individually marginally significant coefficient; the other by-grade coefficients are positive but below conventional significance, which reflects the smaller per-grade samples ($\sim$830 to 1,050 students each) rather than evidence of effect heterogeneity. Section V tests directly whether the effect varies with pre-PAL ability; Section VI examines on which test items the gain materialises.

V. Heterogeneity by Baseline Ability

We test whether the Free Flow effect varies systematically with students' pre-PAL math achievement. Because Free Flow is randomly assigned within school and grade, an interaction with a pre-treatment test-score measure is causally identified by the experimental design.

We use a cohort-specific pre-PAL baseline math score. The Grade 6 endline cohort entered PAL only in AY 2024–25, so we use their AY 2023–24 government Summative Assessment 1 (taken in their Grade 5 year, before PAL). Grades 7–9 had begun PAL in AY 2023–24, so we use their AY 2022–23 Summative

Assessment 1 (taken in their Grade 5–7 year, also pre-PAL). This matches the cohort-specific baseline construction in our companion paper. We then partition students into within-school-by-grade terciles of this baseline; mean baseline scores in the three tercile bins are 17, 30, and 46 respectively, on a 100-point government scale.

Table 5 reports Free Flow effects on θ^{std} at each tercile. Point estimates are positive at all three terciles — $+0.049\ \sigma$ ($p = 0.23$) at the lowest, $+0.060\ \sigma$ ($p = 0.13$) in the middle, and $+0.081\ \sigma$ ($p = 0.11$) at the highest. None of the per-tercile coefficients reaches conventional significance individually; the joint F-test of equal Free Flow effects across terciles is far from rejection ($F = 0.12$, $p = 0.88$): with $\sim$1,150 students per tercile, the data are consistent with a uniform Free Flow effect across the baseline-ability distribution. We find no evidence that lower-baseline students benefit disproportionately from access to deeper remediation, despite the prior that they have "more room to gain" from foundational content.

We complement the tercile analysis with three distributional checks on θ^{std} itself: an F-test for equal SD across arms, its residualised counterpart on within-school-by-grade-by-section deviations, and a quantile regression of θ^{std} on Free Flow (with the §III covariate vector and FE) at $q \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, with cluster-bootstrap SE (200 reps; schools resampled with replacement, $S = 20$). Appendix Table 6 and Appendix Figure 8 report the results. The arm SDs are essentially equal — raw 1.000 versus 0.990 ($F = 1.02$, $p = 0.65$); residualised 0.888 versus 0.900 ($F = 1.03$, $p = 0.54$). Quantile regression localises the gain at the lower-middle of the distribution: the FF coefficient is $+0.083\ \sigma$ at $q = 0.25$ ($p = 0.002$, 95% CI $[+0.022, +0.122]$) and $+0.049\ \sigma$ at the median ($p = 0.07$, $[+0.001, +0.101]$); the $q = 0.75$ point estimate is similar in magnitude ($+0.039\ \sigma$) but not distinguishable from zero ($p = 0.32$), and both tails are indistinguishable from zero ($q = 0.10$: $+0.028\ \sigma$, $p = 0.49$; $q = 0.90$: $+0.033\ \sigma$, $p = 0.52$). The empirical CDFs separate visibly across the lower-to-middle range and converge at the top, matching the quantile pattern. We read this as compatible with the broadly-uniform tercile pattern: the gain is positive across the distribution but slightly larger where students sit somewhat below the within-stratum median rather than at the tails.

TABLE 5—FREE FLOW EFFECT BY PRE-PAL BASELINE MATH TERCILE

	FF effect (1)	(SE) (2)	p-value (3)
FF effect at each baseline tercile (within school × grade)			
Tercile 1 (lowest pre-PAL baseline)	+0.049	(0.041)	0.230
Tercile 2 (middle)	+0.060	(0.040)	0.131
Tercile 3 (highest)	+0.081	(0.051)	0.113
Differential effect (vs. T1)			
Tercile 2 − Tercile 1	+0.010	(0.062)	0.867
Tercile 3 − Tercile 1	+0.032	(0.065)	0.620
Joint F (interactions = 0)	0.12		0.884
Mean pre-PAL baseline (T1, T2, T3)	16.7 / 31.2 / 47.9		
Observations	3,492		
R^2	0.348		

Notes: Free Flow effects on θ^{std} at each within-school-by-grade tercile of the cohort-specific pre-PAL baseline math score (Grade 6: AY 2023–24 SA1; Grades 7–9: AY 2022–23 SA1). Specification: OLS of θ^{std} on Free Flow, two interaction terms (Free Flow × Tercile 2 and Free Flow × Tercile 3), two tercile main effects, the §III covariate vector, and school-by-grade-by-section fixed effects; standard errors clustered at the school level ($S = 20$). Terciles are constructed within school × grade (not within section) to keep bins large. FF-effect rows for Tercile 2 and Tercile 3 are linear combinations of the Free Flow main effect and the corresponding interaction. The joint F-test asks whether the two interactions are jointly zero. 263 students drop from the sample because their pre-PAL baseline test record is missing. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE 6—DISTRIBUTIONAL EFFECTS OF FREE FLOW ON θ^{std}

Panel A: F-test on equal SD across arms (variance ratio)	SD(SQ)	SD(FF)	F	p	
Raw θ^{std}	1.000	0.990	1.021	0.650	
Residualised on FE	0.888	0.900	1.029	0.543	
Panel B: Quantile regression FF coefficient (cluster-bootstrap SE, $S = 20$ schools)	Coef	SE	p	95% CI	N
$q = 0.10$	+0.028	(0.040)	0.488	[-0.035, +0.122]	3,755
$q = 0.25$	+0.083***	(0.027)	0.002	[+0.022, +0.122]	3,755
$q = 0.50$	+0.049*	(0.027)	0.071	[+0.001, +0.101]	3,755
$q = 0.75$	+0.039	(0.039)	0.322	[-0.031, +0.120]	3,755
$q = 0.90$	+0.032	(0.050)	0.516	[-0.061, +0.139]	3,755

Notes: Panel A reports two-sided variance-ratio (Snedecor) F-tests for equality of SD(θ^{std}) across arms. Row 1 uses raw θ^{std}; row 2 first residualises θ^{std} on the school × grade × section fixed effects and tests the within-stratum residual SD. Panel B reports the FF coefficient from quantile regressions of θ^{std} on Free Flow + the §III covariate vector + school × grade × section FE at five quantiles. Standard errors and 95% CIs are pairs cluster-bootstrap (200 replications, schools resampled with replacement, $S = 20$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

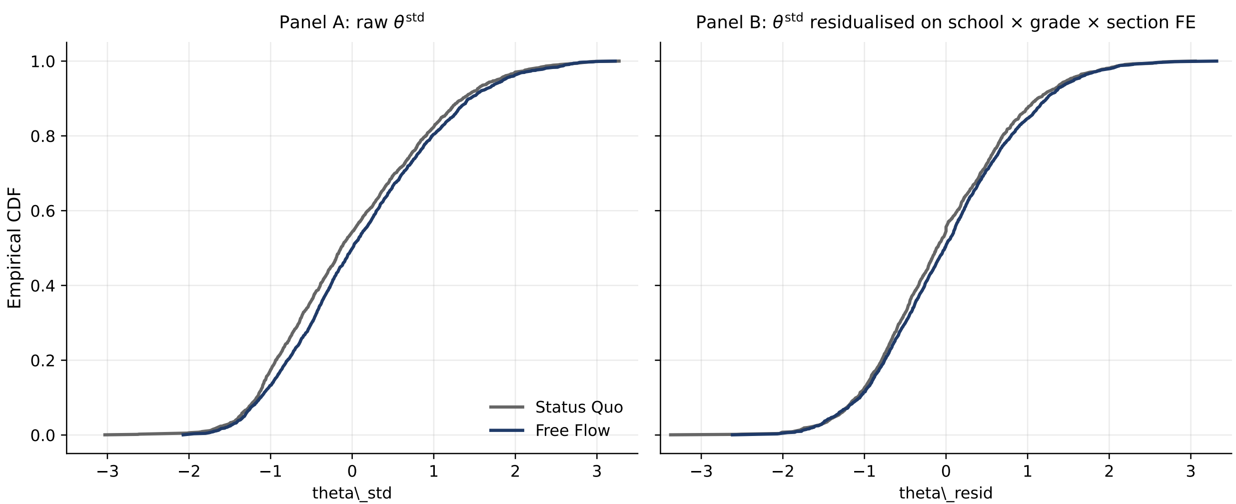

FIGURE 8. EMPIRICAL CDFS OF θ^{std} BY ARM

Notes: Panel A: empirical CDFs of raw θ^{std}, separately for Free Flow (navy) and Status Quo (grey). Panel B: same after residualising θ^{std} on school $\times$ grade $\times$ section fixed effects. The Free Flow CDF lies visibly to the right of Status Quo through the lower-to-middle range and converges at the upper tail, consistent with a treatment effect concentrated below the median rather than a strict mean-shift. Companion to Appendix Table 6.

VI. Where the Learning Lands: Item-Level Decomposition

The aggregate effects in Sections IV and V tell us *how much* Free Flow moves standardised math ability; they do not tell us *where on the curriculum* the gain sits. The endline instrument — a 72-item Math Master Challenge spanning content grades 2 through 9, with each student answering a grade-adaptive subset — lets us ask this directly. If the additional deep-remediation content Free Flow renders is producing foundational learning, the gain should be visible on test items at the grade levels the extra content covers.

For each student we compute three sub-abilities using the same 2PL item parameters from our primary IRT fit, but conditioning each estimate on items in only one content bucket relative to the student's enrolled grade: at-grade items (g), shallow-remediation items ($g-1$ and $g-2$), and deep-remediation items ($g-3$ through $g-5$). Each sub-ability is then standardised within enrolled-grade cohort, so Free Flow coefficients report in SD units of the relevant cohort's distribution of that sub-ability.

Table 7 and Figure 9 report Free Flow effects on each sub-ability, pooled across grades and separately by enrolled grade. Pooling across grades, the Free Flow effect is essentially zero on at-grade items ($+0.016\ \sigma$, $p = 0.56$), strongest on shallow-remediation items ($+0.072\ \sigma$, $p = 0.001$), and intermediate on deep-remediation items ($+0.051\ \sigma$, $p = 0.018$). The shallow-item effect is the largest single coefficient in the paper. A stacked test of equal Free Flow effects across the three buckets returns $F = 1.53$ ($p = 0.242$); the pairwise contrast of shallow versus at-grade effects is $+0.041\ \sigma$ ($p = 0.09$), suggestive but not statistically definitive at conventional levels. The pattern is qualitatively consistent with a foundational-learning mechanism: deeper content built skills that transferred up to test items at content depths just below enrolled grade.

The by-grade decomposition is sharpest at Grade 6. Grade 6 Free Flow students show a $+0.164\ \sigma$ gain ($p = 0.006$) on shallow-remediation items (Grade-4 and Grade-5 content, from the Grade-6 cohort's perspective), a smaller $+0.089\ \sigma$ effect ($p = 0.19$) on deep-remediation items (Grade-2 and Grade-3 content), and no detectable effect on at-grade (Grade-6) items ($+0.066\ \sigma$, $p = 0.40$). Grades 7–9 show smaller and less significant point estimates at every content depth; the strongest Grade 7–9 result is the deep-item coefficient at Grade 7 ($+0.075\ \sigma$, $p = 0.27$). The pooled shallow-item significance is therefore driven primarily by Grade 6 with consistent positive contributions from the other grades.

Two features of the Grade-6 pattern are worth emphasising. First, Grade-6 Free Flow students engage with roughly 200 minutes of $g-3$-and-below material per year — nearly twice the per-year rate of Grade 7–9 Free Flow students — yet the Grade-6 gain on the very items that match this distinctive dose ($+0.09\ \sigma$ on

Table 7—Free Flow effect on endline performance by content grade of test item

	All grades (1)	Grade 6 (2)	Grade 7 (3)	Grade 8 (4)	Grade 9 (5)
At-grade items	+0.016	+0.066	+0.013	-0.056	+0.037
	(0.028)	(0.079)	(0.055)	(0.060)	(0.056)
Shallow-remediation items ($g-1$, $g-2$)	+0.072***	+0.164***	+0.044	+0.031	+0.052
	(0.023)	(0.060)	(0.054)	(0.057)	(0.050)
Deep-remediation items ($g-3$ and below)	+0.051**	+0.089	+0.075	+0.018	+0.029
	(0.022)	(0.068)	(0.068)	(0.054)	(0.067)
Joint F (equal FF effect across buckets, pooled)	$F = 1.53$, $p = 0.242$				
Students	3,755	832	883	984	1,056
Items per bucket (G6/G7/G8/G9)	at: 10 / 10 / 6 / 6; shallow: 20 / 20 / 20 / 16; deep: 20 / 30 / 40 / 50				

Notes: Free Flow effects on IRT sub-abilities computed from the 72-item Math Master Challenge. For each student we compute three sub-abilities using the globally fitted 2PL item parameters and conditioning each estimate on only items in one content bucket relative to the student's enrolled grade: at-grade (g), shallow-remediation ($g-1$, $g-2$), and deep-remediation ($g-3$ through $g-5$). Each sub-ability is standardised within enrolled-grade cohort. Specification: OLS of the sub-ability on Free Flow plus the §III covariate vector and school-by-grade-by-section fixed effects; standard errors clustered at the school level ($S = 20$). The joint F-test asks whether Free Flow effects are equal across the three buckets when all three are stacked into a single student-by-bucket panel and Free Flow is interacted with bucket indicators. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Grade-2 and Grade-3 items) is not statistically detectable. Second, the *largest* measured gain is on items one to two grades *above* the deepest content they engaged with: the gain on Grade-4 and Grade-5 items ($+0.16\ \sigma$) is roughly twice the gain on the Grade-2 and Grade-3 items where the distinctive content was rendered. Three readings are consistent with this spill-up pattern. The deep content may build prerequisite skills — basic arithmetic, place value, whole-number operations — that unlock Grade-4 and Grade-5 content (fraction operations, introductory algebra) on which Grade-6 students sit closer to a competency threshold. Mean percent correct on Grade-6 students' Grade-2–Grade-3 items (around 0.54–0.58) is already high enough that additional gain has less room to register on the easier items; Grade-4–Grade-5 items at 0.45–0.51 have more headroom. And the at-grade Grade-6 items at 0.42 may sit too far above the foundations the experiment delivered for gains to transfer within the available exposure time. The design does not separate these channels, but the pattern — improvement one to two content-grades above where the distinctive dose landed, with no corresponding improvement at enrolled-grade level — is coherent with a finite per-year transfer distance.

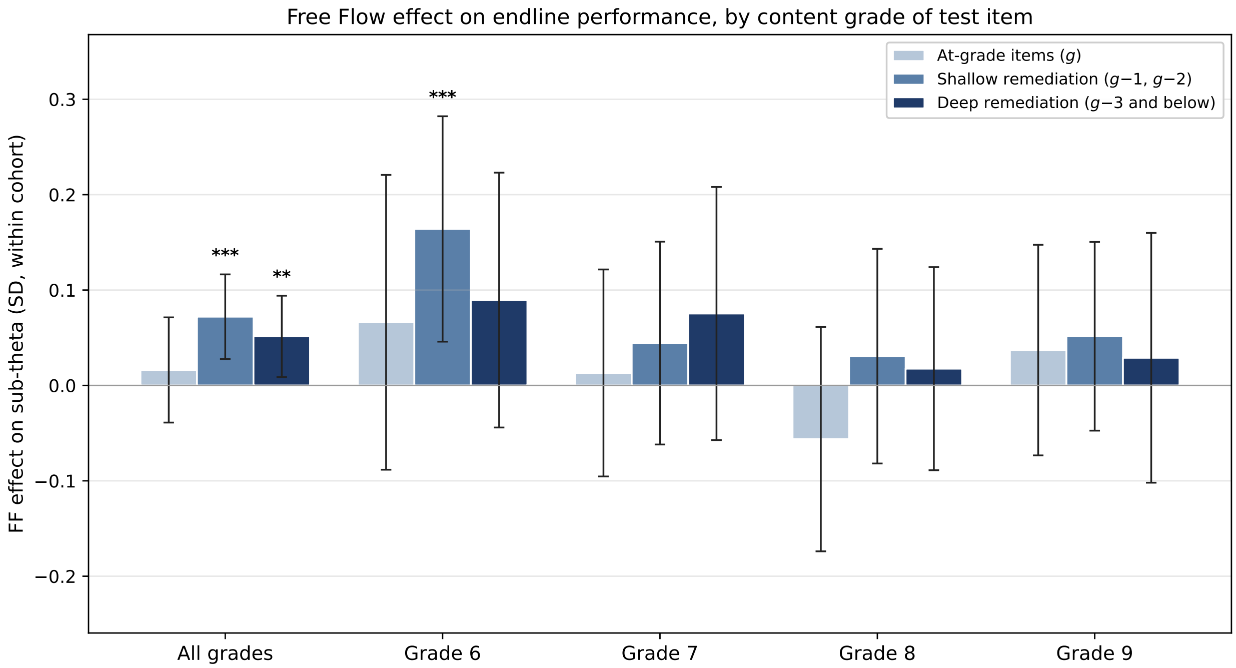

FIGURE 9. WHERE THE FREE FLOW GAIN LIVES: EFFECT BY ENROLLED GRADE AND ITEM CONTENT DEPTH

Notes: Free Flow effect on endline math ability, in SD units of the within-cohort distribution, separately for three item buckets defined by content grade relative to the student's enrolled grade: at-grade (g, light), shallow-remediation ($g-1$ and $g-2$, mid-tone), and deep-remediation ($g-3$ and below, dark). Error bars show 95% confidence intervals from OLS with the §III covariate vector and school-by-grade-by-section fixed effects, standard errors clustered at the school level. Stars above bars report significance levels (* $p<0.10$, ** $p<0.05$, *** $p<0.01$). The Grade 6 cohort, which receives nearly twice the deep-remediation dose of Grades 7–9 (Section II), shows a $+0.16\ \sigma$ gain on shallow-remediation items (Grade-4 and Grade-5 content), a smaller and statistically indeterminate gain on deep-remediation items (Grade-2 and Grade-3 content), and no detectable gain on at-grade items. The pooled shallow-item effect is highly significant ($+0.07\ \sigma$, $p=0.001$); other grade cohorts contribute consistent positive but smaller point estimates.

VII. Discussion

Free Flow's experimental contrast is access to remediation content three to five grade levels below enrolled grade — material that the Status Quo algorithm does not render. The behavioural response to that access is, on average, an additional ≈ 200 minutes of deep-remediation engagement per student over the two-year intervention, making up roughly 90% of the total Free Flow–Status Quo engagement gap of ≈ 218 minutes; the remaining 10% appears as a small Free Flow excess at at-grade content, with shallow remediation essentially balanced. The aggregate learning effect on standardised math ability is small but statistically significant: $+0.053\ \sigma$ ($p=0.016$) in the baseline specification.

These magnitudes imply a per-hour return of approximately $0.02\ \sigma$ per cumulative hour of deep-remediation exposure at the Free Flow–Status Quo margin. This is in the same order of magnitude as the per-hour return to total PAL exposure estimated by the IV strategy in our companion paper ($\approx 0.03\ \sigma$/hour).

The aggregate Free Flow effect is therefore consistent with deep remediation being roughly as productive per minute as PAL's at-grade and shallow-remediation content; the small aggregate effect reflects the small dose differential rather than a low marginal return.

The aggregate is broadly distributed across the cohort. Point estimates are positive at three of four enrolled grades — Grade 6 ($+0.113\ \sigma$, $p = 0.075$), Grade 7 ($+0.059\ \sigma$), Grade 9 ($+0.031\ \sigma$) — and essentially zero at Grade 8 ($+0.016\ \sigma$). The Grade 6 effect is the largest in absolute value and the only individually marginally significant by-grade coefficient; the others are positive but below conventional significance because of smaller per-grade samples ($\sim$830 to 1,050 students). The broad uniformity is itself informative: the pooled effect does not hinge on a single grade-specific outcome. Heterogeneity by pre-PAL baseline ability is also undetectable: the joint F-test of equal Free Flow effects across within-school-by-grade ability terciles is $F = 0.12$ ($p = 0.88$), with positive coefficients at all three terciles ($+0.049$, $+0.060$, $+0.081\ \sigma$) but no monotone gradient in either direction. Within the precision of this experiment, deeper remediation appears to deliver a small, broadly uniform gain on standardised math test scores.

The item-level decomposition in Section VI sharpens *where* the gain materialises. Pooled across grades, the Free Flow effect is largest on shallow-remediation items ($+0.072\ \sigma$, $p = 0.001$) and intermediate on deep-remediation items ($+0.051\ \sigma$, $p = 0.018$); at-grade items show no detectable gain ($+0.016\ \sigma$, $p = 0.56$). The pattern is most distinct at Grade 6, where the shallow-item effect reaches $+0.164\ \sigma$ ($p = 0.006$). That the gain shows up on shallow-remediation items — Grade-4 and Grade-5 content, material both arms' algorithms render — despite the Free Flow contrast being located at $g-3$ and below, is the learning-science signature we take most seriously. It is consistent with deep content building prerequisite skills that transfer *up* to content one or two grades above where the distinctive dose landed. Whether the gain would propagate further — to at-grade items in a second experimental year, or to deeper levels given a longer-running exposure — is a question this experiment cannot answer but that would be informative for remediation-policy design.

A direct test of per-minute returns to deep-remediation engagement is unavailable in this design. Within Free Flow, students who log more deep-remediation minutes are precisely those whom the algorithm diagnoses as furthest behind, and so OLS regressions of θ^{std} on minutes-by-content-depth pick up a strong selection signature in which the deeper-remediation slope appears negative. Our companion paper instruments PAL hours with section-enrolment variation; that strategy is not available here because Free Flow assignment is itself the source of deep-remediation variation.

Two limitations bound the inferences we draw. First, statistical power in this within-school design is concentrated in pooled tests; tercile- and grade-level effects are estimated from $\sim$1,150 and $\sim$830–1,050 students respectively and are individually noisy. Second, the twenty-school sample is drawn from the high-usage tail of Andhra Pradesh's Personalised Adaptive Learning programme; in schools with weaker PAL implementation, both the size of the responding subset and the dose of deep-remediation engagement Free Flow induces may differ.

The aggregate Free Flow effect is small. Whether this reflects a genuinely low marginal return to deeper remediation or a treatment dose too modest to produce a larger detectable effect cannot be distinguished from this experiment alone. A larger or longer-dose experiment would be required to do so.

References